

AI-Based Financial News Sentiment Analysis for Stock Market Prediction

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Abstract— This paper presents an AI-Based Financial News Sentiment Analysis for Stock Market Prediction system that combines financial news with historical stock market data to predict stock price direction. The system uses FinBERT for financial sentiment analysis and extracts 35 engineered features covering sentiment dynamics, market momentum, RSI, MACD, volatility, volume behavior, and sentiment-market interactions. These features are integrated using the proposed FinSent-Momentum Fusion (FSMF) framework and used with a Gradient Boosting classifier to predict whether the stock market will move UP or DOWN. The system is evaluated on the TRACE AAPL dataset, where the proposed approach achieves 69.12% accuracy on the held-out test period. An interactive Streamlit dashboard further provides sentiment analysis and stock movement predictions to support data-driven financial decision-making.

Keywords— Financial News, Sentiment Analysis, FinBERT, Stock Market Prediction, NLP, Feature Engineering, Gradient Boosting, FSMF

I. INTRODUCTION

Financial news plays an important role in influencing investor sentiment and stock market movements. Large volumes of financial news are published every day, making manual analysis difficult, time-consuming, and challenging to perform consistently. News articles can contain positive, negative, or neutral information that may influence market trends. Natural Language Processing (NLP) and machine learning techniques can analyze this information and identify meaningful sentiment patterns.

This project proposes an AI-Based Financial News Sentiment Analysis for Stock Market Prediction system that combines financial news analysis with historical stock market data. The system uses FinBERT to extract positive, negative, and neutral sentiment from financial news and applies feature engineering to generate 35 features involving sentiment dynamics, market momentum, RSI, MACD, volatility, volume behavior, and sentiment-market interactions.

The proposed FinSent-Momentum Fusion (FSMF) framework combines these features and uses a Gradient Boosting classifier to predict stock movement as UP or DOWN. An interactive Streamlit dashboard is also developed to display sentiment analysis and prediction results.

The system aims to support data-driven financial analysis by combining textual news information with historical market behavior.

II. RELATED WORK

Several studies have explored the relationship between financial news, investor sentiment, and stock market movements. Traditional approaches generally relied on statistical techniques and manually engineered features, while recent approaches increasingly use machine learning, deep learning, and transformer-based language models.

Ding et al. investigated event-driven stock prediction using deep learning and demonstrated the importance of incorporating information extracted from news events into stock prediction models.

Araci introduced FinBERT, a transformer-based language model specifically designed for financial sentiment analysis.

FinBERT provides sentiment classifications that are better suited to financial language than general-purpose sentiment analysis models.

Xu and Cohen studied stock movement prediction using social media information together with historical prices, demonstrating the usefulness of combining textual information with numerical market information.

Recent research has also explored transformer-based approaches and sentiment analysis for stock trend prediction. However, many existing approaches focus primarily on sentiment scores or combine textual and numerical information without explicitly integrating multiple technical market indicators.

III. PROPOSED SYSTEM

The proposed system combines financial news sentiment with historical stock market information to predict stock price direction.

The complete workflow consists of the following stages:

1. Financial news and historical AAPL market data are collected.
2. The news data are cleaned and aligned with market dates.
3. FinBERT is used to obtain positive, negative, and neutral sentiment scores.
4. Multiple market and sentiment features are engineered.
5. The 35 features are integrated using the proposed FSMF framework.
6. Gradient Boosting is trained for UP/DOWN prediction.
7. The trained model is integrated into a Streamlit application.

IV. METHODOLOGY

A. Data Collection

The system uses financial news from the TRACE ACL18 AAPL dataset along with historical AAPL stock market

data. The financial news is associated with corresponding dates so that textual sentiment can be analyzed together with market behavior.

B. Data Preprocessing

The collected data undergo preprocessing before model training. News text is cleaned and processed for sentiment analysis. Historical market data are organized chronologically, and financial news is aligned with the corresponding trading dates.

The target variable represents the next stock movement:

$$Y_t = \begin{cases} 1, & P_{t+1} > P_t \\ 0, & P_{t+1} \leq P_t \end{cases} \quad (1)$$

where P_t represents the stock price at time t .

C. Financial Sentiment Analysis

FinBERT is used to analyze the financial news text. For each news item, the model generates sentiment probabilities corresponding to positive, negative, and neutral classes.

The sentiment representation can be expressed as:

$$S_t = [S_{positive}, S_{negative}, S_{neutral}] \quad (2)$$

These sentiment values are subsequently incorporated into the feature engineering stage.

D. Feature Engineering

A total of 35 engineered features are used in the proposed system.

The features represent different aspects of financial information, including:

- Sentiment features
- Sentiment dynamics
- Market momentum
- RSI
- MACD
- Volatility
- Volume behavior
- Sentiment-market interactions

These features provide the model with both textual and numerical information.

E. FinSent-Momentum Fusion

The proposed FinSent-Momentum Fusion (FSMF) framework combines financial sentiment information with market-based features.

The general feature representation is:

$$F_t = [S_t, M_t, T_t, V_t, Vol_t, I_t] \quad (3)$$

where:

- S_t = sentiment features
- M_t = market momentum features
- T_t = technical indicator features

- V_t = volatility features
- Vol_t = volume-related features
- I_t = sentiment-market interaction features

The resulting feature representation is provided to the machine learning classifier.

F. Gradient Boosting Classification

Gradient Boosting is used as the final classification model.

The classifier predicts one of two classes:

- 1 – UP
- 0 – DOWN

The model learns relationships between the engineered features and subsequent stock movement.

V. EXPERIMENTAL SETUP

The system is evaluated using the TRACE AAPL dataset and historical AAPL market information.

The baseline approach uses Logistic Regression with basic sentiment and market features. The proposed approach uses the 35 engineered features through the FSMF framework and Gradient Boosting classifier.

Table 1: Model Performance

Model	Features	Accuracy
Logistic Regression	Basic sentiment and market features	56%
FSMF + Gradient Boosting	35 engineered features	69.12%

VI. RESULTS AND DISCUSSION

The baseline Logistic Regression model achieved an accuracy of 56% on the test data.

The proposed FSMF framework combined with Gradient Boosting achieved an accuracy of 69.12% on the held-out TRACE AAPL test period.

This represents an improvement of 13.12 percentage points over the baseline.

Table 2: Feature Categories

Category	Purpose
Sentiment	News polarity information
Sentiment Dynamics	Changes in sentiment
Momentum	Market movement
Technical Indicators	RSI and MACD
Volatility	Market risk behavior
Volume	Trading activity
Interactions	Sentiment-market relationships

The results indicate that combining financial sentiment with multiple market characteristics provides a more informative representation than relying only on basic sentiment information.

VII. STREAMLIT APPLICATION

An interactive Streamlit application was developed to provide a simple interface for users.

The user enters financial news into the application. The news is processed by FinBERT to obtain sentiment scores. The engineered feature pipeline then generates the required model features, which are passed to the trained Gradient Boosting classifier.

The application displays:

- Input financial news
- Positive sentiment score
- Negative sentiment score
- Neutral sentiment score
- Predicted stock movement

The final prediction is displayed as either UP or DOWN.

VIII. LIMITATIONS AND FUTURE SCOPE

The current system is evaluated using AAPL-related data and therefore may not directly generalize to all stocks or market conditions.

The prediction accuracy also depends on the quality and availability of financial news and historical market data.

Future work can extend the system to multiple stocks, additional financial news sources, and larger datasets. More advanced transformer-based models and time-series architectures can also be explored.

The Streamlit application can additionally be extended with real-time news collection, live market data, portfolio-level analysis, and improved visualization.

IX. CONCLUSION

This paper presented an AI-Based Financial News Sentiment Analysis for Stock Market Prediction system that combines financial news with historical stock market information.

FinBERT is used to extract financial sentiment, while 35 engineered features capture sentiment dynamics, market momentum, technical indicators, volatility, volume behavior, and sentiment-market interactions. These features are integrated using the proposed FinSent-Momentum Fusion framework and classified using Gradient Boosting.

The proposed approach achieved 69.12% accuracy on the held-out TRACE AAPL test period compared with 56% for the baseline Logistic Regression model.

The developed Streamlit application provides an interactive interface for performing financial news sentiment analysis and stock movement prediction.

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